SPECTRA: A Traceability Ontology for the 3GPP RAN Standardization Process

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Abstract. The 3GPP standardization process produces a large document ecosystem across five Radio Access Network (RAN) Working Groups; to the best of our knowledge, no publicly available OWL ontology specifically targets its document lifecycle and multi-hop traceability process. We present **SPECTRA** (*SPECification TRAcing*), a reusable OWL ontology organized around four concerns: contributions, resolutions, specifications, and cross-group communication. SPECTRA is grounded in 137 competency questions elicited from practicing standards engineers; representative CQs with executable Cypher translations and a smaller set of SPARQL examples accompany the release. The ontology is validated by a cumulative CQ protocol across five incremental development phases, and its applicability is demonstrated by cross-WG instantiation across RAN1–RAN5 (the RAN1-designed schema required no additional classes, one additional object property, and two genuinely new data properties to cover RAN2–RAN5). SPECTRA currently serves as the semantic layer of a standards-engineering knowledge infrastructure at Samsung System LSI.

Keywords: Ontology engineering · Knowledge graph · 3GPP · Competency questions · Telecom standardization

1 Introduction

The 3rd Generation Partnership Project (3GPP) is the global standards body responsible for defining mobile communication technologies, from 4G LTE and 5G NR to the upcoming 6G standards. The standardization process is contribution-driven, with member companies submitting technical documents through 3GPP's Working Group (WG) structure. The RAN (Radio Access Network) Technical Specification Group alone encompasses five WGs (RAN1–RAN5), each responsible for a distinct layer of the radio interface — from physical layer procedures to conformance testing.

3GPP specifications are the normative reference that commercial 5G implementations and conformance tests must align with. Yet the corpus spans hundreds of normative specifications and evolves through a process that produces hundreds of thousands of Working Group documents. Acquiring even par-

tial working knowledge typically requires substantial domain expertise and prolonged immersion, creating a barrier that compounds along several axes: *cross-WG* (a RAN1 decision propagates to RAN2/3/4 through Liaison Statements few engineers track end-to-end), *cross-organization* (each company has internal context only for its own contributions), and *industry–academia* (researchers see the published specifications but not the meeting-level decisions that shaped them). The traceability path TDoc → Resolution → CR → TS Section → Spec, which captures *why* a clause reads as it does and where it resides, currently exists as implicit institutional knowledge and is typically reconstructed by hand — error-prone and time-consuming.

To the best of our knowledge, no publicly available OWL ontology specifically targets the 3GPP RAN standardization document lifecycle. Recent 3GPP-domain resources — TSpec-LLM [9] for retrieval-augmented access to specification text and *telecom-kg-rel19* [18] for spec-text concept extraction — model the *content* of specifications but are not designed to directly represent the standardization *process* that produced them — e.g., which TDocs led to a clause, which CRs were triggered by a Liaison Statement, or which Working Assumptions were promoted to Agreements.

In this paper, we present **SPECTRA** (*SPECification TRAcing*), a reusable OWL ontology for modeling the types, relationships, and lifecycle of 3GPP RAN standardization documents. The contributions are:

1. **The SPECTRA OWL ontology** — a reusable OWL resource for 3GPP RAN standardization documents and their lifecycle, comprising 26 classes and 134 properties (53 object properties, 81 data properties) organized around contributions, resolutions, specifications, and cross-group communication; the ontology reuses Dublin Core and FOAF and is licensed CC-BY 4.0 (release details in Section 7).
2. **A competency-question grounding** — SPECTRA's classes, properties, and axioms are derived from 137 competency questions elicited from practicing 3GPP standards engineers at Samsung System LSI; representative CQs with executable Cypher translations and a smaller set of SPARQL examples accompany the public release for inspection and reuse without access to the 3GPP source corpus; the complete suite is retained as internal validation material.
3. **Cross-WG applicability and instantiation evidence** — the schema has been instantiated at realistic scale across all five RAN Working Groups (RAN1–RAN5), and applying the RAN1-designed schema to RAN2–RAN5 required *no additional classes, one additional object property, and two genuinely new data properties* (Table 6). The instantiated knowledge graphs and operational parsing pipeline are not redistributed because the underlying TDocs are subject to 3GPP copyright.
4. **A cumulative CQ validation protocol** — prior CQ-driven methodologies (LOT [12], SAMOD) typically apply CQ validation within a single design iteration; we instead develop SPECTRA through five incremental phases (25, 34, 45, 15, 18 CQs; 137 total) and re-validate the accumulated suite at

each phase transition. Three quality-assurance cases surfaced by this protocol were resolved before release.

Sections 2–8 cover related work, 3GPP background, ontology design, the CQ-driven cumulative validation methodology, evaluation, availability/FAIRness, and conclusion respectively.

2 Related Work

2.1 Ontologies for Domain Knowledge Management

Ontologies have been widely adopted as the semantic backbone for domain-specific knowledge management systems. In cultural heritage, the Polifonia Ontology Network [2] provides 15 modular ontologies for musical heritage, developed through eXtreme Design with 361 competency questions from 10 pilots. In the energy domain, the Hydrogen Ontology (HOLY) [1] models the structure of hydrogen-based markets through CQ-driven design. In industry, Cao et al. [4] describe a knowledge graph for the quality management of Bosch products. For technical standards, Grangel-González and Vidal [6] use the Standards Ontology (STO) as the schema for an Industry 4.0 Knowledge Graph covering more than 200 industrial standards, but their focus is on *inter-standard* relationships rather than the *intra-standard* document structure central to our work. In telecommunications, NORIA-O [16] targets anomaly detection and incident management in ICT systems rather than the standardization process itself. The most directly comparable recent effort in the 3GPP domain is *telecom-kg-rel19* [18], a knowledge graph extracted from Release-19 specification *text* for retrieval-augmented LLM reasoning; SPECTRA differs in that it models the standardization *process and document lifecycle* (TDoc, Resolution, CR, LS, TR, Section, etc.) rather than concept structures embedded in spec text. To the best of our knowledge, no publicly available OWL ontology specifically targets the 3GPP RAN standardization document lifecycle and traceability process. Table 1 positions SPECTRA against the closest comparable resources along the axes most relevant to the ISWC Resource Track.

2.2 Ontology Engineering Methodologies

The Ontology 101 guide [10] established a widely-adopted methodology with competency questions (CQs) as the primary requirements mechanism [7]. LOT [12] extends this for industrial settings with FAIR compliance, the NeOn methodology [15] provides scenario-based pathways for building ontology networks (including reuse of existing ontological and non-ontological resources), and SAMOD [11] draws on Test-Driven Development principles. While these methodologies support iterative refinement and, in NeOn's case, modular network development, they do not explicitly frame CQ validation as a *cumulative* protocol across multiple development phases — extending an ontology for a new sub-domain while re-executing the full CQ suite accumulated from prior phases,

Table 1. Differentiation of SPECTRA against closely related Semantic Web resources along axes most relevant to the ISWC Resource Track, based on each work's published abstract / availability statement. *Process* = explicitly models a domain process/lifecycle. *Multi-hop trace* = supports backward traceability over ≥ 3 class-types. *CQ-grounded* = design driven by an explicit CQ set. *Cross-subdomain validation* = the released schema validated across ≥ 2 sub-domains.

Resource	Domain	Process	Multi-hop trace	CQ-grounded	Cross-subdom. valid.	Real-world deploy.
Polifonia [2]	cultural heritage	limited	no	yes (361 CQs)	yes (15 modules)	multiple stakeholders
HOLY [1]	hydrogen market	no	no	yes (test cases)	no	Atlant-H project
NORIA-O [16]	ICT incidents	yes (incident)	limited	no	no	telco operator
Grangel & Vidal [6]	I40 standards	no	no	no	no	I40KG
TSpec-LLM [9]	3GPP text	no	no	no	no	LLM benchmark
telecom-kg-rel19 [18]	3GPP Rel-19 text	no	no	no	no	LLM-RAG eval
SPECTRA (ours)	**3GPP RAN process**	**yes**	**yes (5-hop)**	**yes (137 CQs)**	**yes (5 WGs)**	Samsung System LSI

SPECTRA's distinguishing claim is the simultaneous presence of process modelling, multi-hop traceability, CQ-grounding, and cross-subdomain (cross-WG) validation; no listed prior resource exhibits all four.

so that no earlier competency is silently regressed. SPECTRA's five-phase cumulative CQ validation (Section 5) uses this protocol as a quality-assurance mechanism for the released resource rather than a replacement for existing methodologies.

In terms of CQ scale, the Polifonia Ontology Network [2] releases a 361-CQ dataset (*PolifoniaCQ*) collected across the network's modules, and Wisniewski et al. [17] analyzed 234 CQs from 5 ontologies and translated them into 131 SPARQL-OWL queries. SPECTRA contributes 137 CQs applied cumulatively to a single cohesive ontology covering five document domains.

2.3 LLM-assisted Ontology Engineering

Large language models have recently been explored as ontology-engineering assistants; Saeedizade and Blomqvist [14] report that GPT-4 reaches "novice human modeler" quality for CQ-driven OWL generation on small general-purpose ontologies. Systematic comparison against LLM-generated counterparts for specialized industrial domains is left to future work.

3 3GPP Standardization Background

3.1 Document Types and Lifecycle

The 3GPP standardization process produces several distinct document types. **Technical Documents (TDocs)** are the primary unit of contribution, carrying

structured metadata including a unique identifier (e.g., R1-2400001), submitting company, target agenda item, and purpose. **Resolutions** record meeting decisions as Agreements, Conclusions, or Working Assumptions. **Change Requests (CRs)** formally propose modifications to specifications. **Technical Specifications (TSs)** are normative documents (e.g., TS 38.211), organized hierarchically into sections. **Technical Reports (TRs)** document feasibility studies. **Liaison Statements (LSs)** are official cross-WG communications.

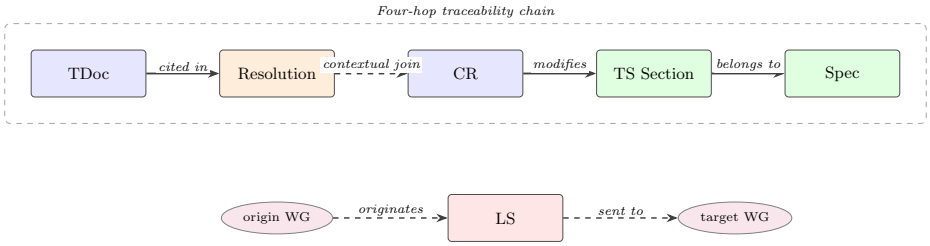

Fig. 1. Document lifecycle in 3GPP RAN. The top row is a reader-facing traceability path from a TDoc contribution to the affected specification: `references` is formally defined Resolution → Tdoc and is drawn in the reading direction, `modifiesSection` links CRs to Sections, and `belongsToSpec` links Sections to Specs. The dashed Resolution–CR step denotes an indirect join through meeting/agenda and CR-pack context rather than a single asserted ontology property. The bottom row summarizes LS-based cross-WG communication via `originatedFrom` and `sentTo`.

3.2 Working Group Structure

The RAN TSG is divided into five Working Groups:

WG	Responsibility	Example Specs
RAN1	Physical layer (PHY)	TS 38.211–214
RAN2	Layer 2/3 protocols	TS 38.331, 38.321
RAN3	Architecture and interfaces	TS 38.401, 38.423
RAN4	Radio performance and RF	TS 38.101, 38.104
RAN5	Conformance testing	TS 38.521, 38.533

Changes in one WG frequently trigger Liaison Statements to other WGs, creating cross-group dependencies. RAN WG meetings convene roughly every two months, with each meeting typically generating thousands of TDocs.

3.3 Concepts an Ontology Must Capture

Three features of the corpus shape the ontology: **scale** (any single WG already accumulates on the order of 10^5 TDocs over its meeting history, so the schema must stay compact while admitting large-scale instantiation), **document lineage** (the TDoc→Resolution→CR→Section→Spec path demands a first-class entity and directed links for each document type), and **cross-group dependencies** (Liaison Statements must be modeled as directed artefacts between Working Groups rather than as free-form text). SPECTRA's class and property layout (Section 4) is organized around these three features.

4 SPECTRA Ontology Design

4.1 Competency Question Analysis

The design of SPECTRA is driven by 137 competency questions collected across five development phases, in collaboration with domain experts — standardization engineers at Samsung System LSI who implement 3GPP specifications in commercial chipsets. While the full text of every CQ and its executable query is retained as internal validation evidence, an **anonymized index of all 137 CQs** (identifier, phase, category, and schema area exercised) is included in the public release as `cqs/cq_index.md`, so that third parties can verify the breadth of the CQ set and the ontology classes/properties each CQ exercises.

CQs are distributed across the development phases as: P1 TDoc metadata (25, e.g. "What TDocs did Company X submit?"), P2 Meeting resolutions (34, "What agreements on Agenda Item Z?"), P3 TS structure (45, "Which CRs modified section S?"), P4 CR documents (15, "What is the reason for change of CR C?"), P5 Technical reports (18, "Which TRs have impacted TS 38.211?"), totalling 137.

4.2 Core Schema

Based on the CQ analysis, we designed SPECTRA following the Ontology 101 methodology [10]. The ontology is encoded in OWL 2 [8] and comprises 26 classes, 53 object properties, and 81 data properties, organized around four conceptual groups (Figure 2).

Document: `Tdoc` with four subclasses (`CR`, `LS`, `Summary`, `SessionNotes`); `CR` → `Spec` via `modifies`; `LS` → `WorkingGroup` via `sentTo/originatedFrom`. **Resolution**: subclasses `Agreement`, `Conclusion`, `WorkingAssumption` (with `promotedTo` for the Agreement promotion). **Specification**: `Spec` (TS) and `Section` forming a tree via `hasSubSection/parentSection` (irreflexive, asymmetric); `TechnicalReport` links to influenced Specs through a reified `TRImpact` so a single TR can affect multiple Specs with per-impact metadata. **Organizational**: `Meeting`, `Company` ($\sqsubseteq$ `foaf:Organization`), `Contact` ($\sqsubseteq$ `foaf:Person`), `WorkItem`, `AgendaItem`, `Release`, `WorkingGroup`.

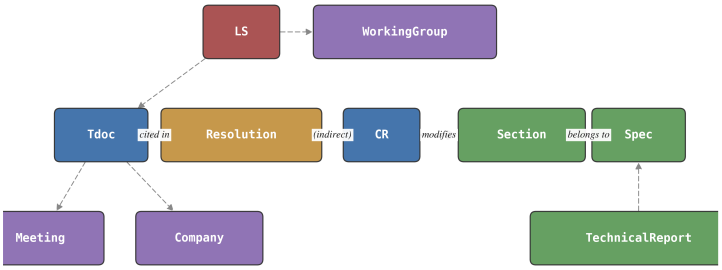

Fig. 2. SPECTRA core schema overview (reader-facing labels in the figure; formal object-property names below). **Central row**: the traceability spine `Tdoc` → `Resolution` → `CR` → `Section` → `Spec`. "*cited in*" = `references` (defined Resolution → Tdoc, drawn reversed); "*modifies*" = `modifiesSection`; "*belongs to*" = `belongsToSpec`. The Resolution → CR arrow is drawn lightly because the released ontology does not declare a direct property between these classes; the transition is reconstructed by joining `resolutionBelongsTo` (Resolution → AgendaItem) with `belongsToCRPack` (CR → CRPack) at the same meeting. **Dashed grey arrows** (unlabeled for clarity) carry secondary relations: `LS` → `Tdoc` is `rdfs:subClassOf`; `LS` → `WorkingGroup` is `sentTo`; `Tdoc` → `Meeting` is `presentedAt`; `Tdoc` → `Company` is `submittedBy`; `TechnicalReport` → `Spec` is `impactsSpec` (mediated by the reified `TRImpact` class). The released ontology declares 26 classes and 134 properties; remaining classes (other `Tdoc`/`Resolution` subclasses, `Contact`, `WorkItem`, `AgendaItem`, `Release`) and FOAF reuse are described in Section 4.

Key Design Decisions. **D1:** Document types are OWL subclasses (not property values) so that type-specific properties can be declared per subclass. **D2:** 23 object properties carry `owl:FunctionalProperty` for an *at-most-one* contract (e.g., a TDoc is presented at at most one meeting); the loader uses these for ingest-time violation detection and also checks minimum-cardinality, which is not enforced at the OWL level. **D3:** The spine (Figure 2) is realized by three direct OPs (`references`, `modifiesSection`, `belongsToSpec`) and a multi-hop join (Resolution → CR via `resolutionBelongsTo`/`belongsToCRPack`); chain-level OPs are deliberately avoided so each declared property keeps a single, verifiable semantic.

4.3 Reuse of Existing Vocabularies

SPECTRA reuses established Semantic Web vocabularies where they naturally apply: **Dublin Core** [5] (`dc:title`, `description`, `creator`, `date`) for ontology/instance metadata; **FOAF** [3] (`Person`, `Organization`) as parent classes for `Contact` and `Company`; **OWL 2** axioms (`Class`, `FunctionalProperty`, `InverseFunctionalProperty`, `inverseOf`, `IrreflexiveProperty`, `AsymmetricProperty`); and **RDFS** (`subClassOf`, `label`, `comment`). Domain-specific classes are introduced only where no suitable external term exists.

We considered but did *not* adopt broader vocabularies as core schema: **PROV-O** would lose 3GPP-specific semantics (Agreement vs Working Assumption, CR-to-Section) unless wrapped in domain subclasses that re-introduce SPECTRA's own classes; **ORG** imposes Organization/OrganizationalUnit/Membership structure beyond what the CQs require; **SKOS** fits hierarchical term taxonomies, not typed first-class entities; **DCAT/VoID** target dataset description rather than document-process modeling. We therefore reuse only DC and FOAF and keep the domain layer self-contained; future alignment with PROV-O via `owl:equivalentClass/equivalentProperty` is possible without changing the released schema.

4.4 Constraints and Axioms

SPECTRA includes 23 `owl:FunctionalProperty` (at-most-one cardinality); 2 `owl:InverseFunctionalProperty` (`hasCR`, `hasTRImpact`); 15 inverse property pairs; 6 `owl:IrreflexiveProperty` covering the section hierarchy, intra-section references, and TR cross-references; and 2 `owl:AsymmetricProperty` on the section hierarchy. These exclude self-loops and 2-cycles in the affected relations. **We deliberately separate semantic commitments encoded in OWL from operational constraints checked by SHACL or the loader**: SHACL shapes (released as `shapes/spectra-core.shacl.ttl`) capture cardinality and class-range constraints reproducibly across triple-store and SHACL-aware tooling, while longer-cycle prevention and minimum-cardinality enforcement remain at the loader level where they require domain-specific recovery semantics.

5 CQ-Driven Cumulative Validation

The cumulative CQ protocol introduced in this section is the quality-assurance mechanism behind the released ontology: each new phase must leave every previously validated competency intact, so that schema refinements are caught at the point of introduction rather than at release.

5.1 Five-Phase Progressive Construction

SPECTRA was developed through five incremental phases, each extending the ontology to cover a new document type while re-executing the full CQ suite accumulated from prior phases. The ontology grew from 11 classes (Phase 1–2, TDoc metadata) to 26 classes (Phase 5, Technical Reports), with the final v1.0.0 release incorporating refactoring identified through cross-phase integrity verification.

5.2 Quality-Assurance Cases

Rerunning the accumulated CQ suite at each phase transition surfaced concrete schema issues. Three representative cases (each traceable via git history):

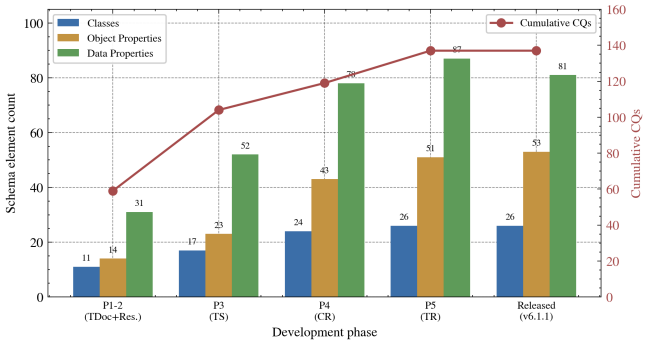

Fig. 3. Schema growth across the five-phase development of SPECTRA, measured from git snapshots at each phase-closing commit. Cumulative CQs (right axis) reach 137 by Phase 5. The data-property drop between Phase 5 and the released version reflects refactoring described in Section 5.

Case 1 (Phase-3 transition) — a relationship-naming inconsistency (`MADEAT` vs `MADE_AT`); three relationship and three property names were harmonized. **Case 2** (Phase-3 audit) — a meeting-ID inconsistency affecting referential integrity; three targeted patches resolved it without schema change. **Case 3** (Phase-3→Phase-4) — three schema refinements: (i) `releaseId` removed as redundant with `releaseName`, (ii) `hasCR` received an `owl:InverseFunctionalProperty` declaration, (iii) the base class `Decision` was renamed `Resolution` to align with 3GPP terminology. The cumulative protocol thus functions as a gating mechanism — gaps a single-phase validation might miss are caught by re-execution against earlier CQ sets.

5.3 Cumulative Regression Verification

At each phase transition, all CQs from Phases 1 through N are re-validated against the schema state introduced by Phase N, executed over an internally instantiated KG that conforms to the schema. The 137 CQs accumulated by Phase 5 all return correct, non-empty results against this internal KG (Section 6, Table 2); after the QA cases above were resolved, our internal validation records did not surface further regressions at subsequent phase transitions (an internal observation, since the instantiated KG is not part of the public release).

6 Evaluation

We organise the evaluation around six dimensions appropriate to a copyright-aware Resource Track artifact: (E1) requirements coverage by CQs, (E2) structural quality of the schema, (E3) scalability evidence at instantiation, (E4) cross-WG generality of the schema, (E5) reusability evidence via public artifacts, and (E6) the rationale of the copyright-aware release design itself (Section 7).

6.1 E1 — Requirements coverage by CQs

All 137 CQs (137/137) return non-empty results against the internally instantiated KG; correctness was confirmed by the standardization engineers who elicited the CQs, comparing the query output against the originating CQ intent. This reflects multiple refinement cycles (Section 5); coverage on third-party instantiations has not been measured. The 137 CQs collectively exercise 20/26 classes and 31/53 OPs as captured by their representative Cypher patterns (matrix in `validation/cq_coverage.json`; category-by-phase distribution in `diagrams/cq_distribution.png`). Unexercised elements include auxiliary content classes (`Chart`, `Figure`, `Table`, `Summary`, `SessionNotes`, `LS`) and OPs reached via inverse paths or join patterns rather than direct edge labels (e.g., `modifiesSection`, `promotedTo`).

Table 2. CQ validation results across ontology phases.

Phase	Domain	CQs	PASS	FAIL	Rate
1	TDoc metadata	25	25	0	100%
2	Meeting resolutions	34	34	0	100%
3	TS structure	45	45	0	100%
4	CR documents	15	15	0	100%
5	Technical reports	18	18	0	100%
	Total	**137**	**137**	**0**	**100%**

6.2 E2 — Structural quality

We report structural metrics of the released ontology (Table 3) and the findings of an automated pitfall scan (Table 4).

Table 3. Ontology structural metrics.

Metric	Value
Classes	26
Object Properties	53
Data Properties	81
Functional Properties	23
Inverse Functional Properties	2
Inverse Property Pairs	15
Irreflexive Properties	6
Asymmetric Properties	2
Reused vocabulary terms	8 (5 DC + 1 DCTERMS + 2 FOAF)

We evaluated SPECTRA with OOPS! [13] via its REST API. The scan reports **no critical pitfalls**, two *Important*, and two *Minor* (Table 4). Both *Important* findings are expected consequences of design decisions: **P34** flags `foaf:Person`/`foaf:Organization` (we reuse FOAF without redeclaring, the recommended pattern), and **P10** flags missing `owl:disjointWith` between document-type classes (a conservative choice because the released CQs do not rely on disjointness, and adding it without instance-level verification risks reasoner inconsistencies). The two *Minor* pitfalls — P08 (annotations on reused FOAF classes; handled by the source vocabulary) and P13 (inverses declared only for bidirectionally queried pairs) — reflect the same reuse-and-parsimony posture.

Table 4. OOPS! results for the released SPECTRA ontology.

Code	Name	Severity	# affected
P08	Missing annotations (on reused FOAF classes)	Minor	2
P10	Missing disjointness (between document-type classes)	Important	1
P13	Inverse relationships not explicitly declared	Minor	23
P34	Untyped class (reused FOAF classes)	Important	2

The release also ships a PyLODE-rendered HTML of the ontology and a SHACL shapes file (`spectra-core.shacl.ttl`) covering core cardinality/range constraints, validated against the bundled synthetic instantiation via pySHACL.

6.3 E3 — Scalability evidence at instantiation

To provide internal evidence that the schema tolerates realistic document volumes, we instantiated SPECTRA against RAN1 data. The graph contains 123,677 Technical Documents from 60 meetings, together with all associated Resolutions, Change Requests, Liaison Statements, Specifications, and Technical Reports (Table 5). Referential integrity for the `submittedBy` relation, the most heavily relied on by the competency questions, is 99.36% (1,000 missing links out of 155,525 `Tdoc`–`Company` associations); the residual gap is traceable to missing source metadata in the original 3GPP TDoc list rather than to schema inadequacy. This instantiation is maintained as internal validation evidence; per Section 7 it is not part of the public release because the underlying TDocs are subject to 3GPP copyright.

6.4 E4 — Cross-WG generality of the schema

A schema designed for one WG can fail silently when applied to a sibling WG with different document conventions. We applied SPECTRA — originally designed from RAN1 CQs — to RAN2–RAN5 without redesign and measured a stricter quantity than label overlap: how many *additional* schema elements each

Table 5. Per-class instance counts in the RAN1 instantiation of SPECTRA (internal validation evidence).

Class	#	Class	#	Class	#
Tdoc	123,677	Resolution	24,710	Section	1,003
CR	10,715	Agreement	21,058	TSTable	680
LS	6,343	Conclusion	2,703	TSFigure	13
Summary	3,468	WorkingAssump.	949	CRPack	501
SessionNotes	678	Meeting	60	TechnicalReport	26
Spec	83	Company	222	TRImpact	29
Release	13	Contact	1,000	Figure	120
WorkingGroup	116	WorkItem	423	Table	1,441
AgendaItem	6,439			Chart	1

WG needed beyond what the RAN1-designed ontology already defined. Table 6 reports the schema-level diff computed from each WG's instantiated KG.

Table 6. Schema diff of RAN2–RAN5 instantiations against the RAN1-designed SPECTRA schema. "WG-specific extensions" counts schema elements present in a non-RAN1 WG graph but not declared in the released ontology (union over RAN2–RAN5).

Schema element	In released SPECTRA	WG-specific extensions	Extension rate
Classes	26	0	0.0%
Object Properties	53	1	1.9%
Data Properties	81	2 †	2.5%

† excluding 20 loader-local properties (17 underscore-prefixed mirror slots and 3 internal scaffolding properties).

The 26 classes cover every document-type distinction the per-WG loaders produced (**zero** additional classes); only one additional object property (`REFERENCES_SPEC`) appears across all four non-RAN1 WGs. The 22 apparent data-property extensions decompose into 20 loader-local attributes (17 underscore-prefixed mirror slots, e.g., `_submittedBy`; 3 scaffolding properties: `id`, `release`, `for_` — none of which represent ontology-level commitments) and two genuinely new data properties (`discussionText`, `summaryId`). **Per-WG ontology-level extensions** (classes / OPs / genuine DPs): RAN2 *0/1/0*, RAN3 *0/1/1* (`summaryId`), RAN4 *0/1/0*, RAN5 *0/1/1* (`discussionText`); the single OP (`REFERENCES_SPEC`) is shared across all four. The total additional schema surface across four sibling WGs is therefore one OP and two DPs on top of the 160 elements already declared — evidence that the RAN1 schema is stable under cross-WG application within the RAN TSG rather than a universal transferability claim.

Conversely, 6 of the 26 RAN1 classes are absent from at least one sibling WG's instantiation. **Per-WG class coverage** (released as `validation/per_wg_class_coverage.json`): RAN1 26/26, RAN2 22/26, RAN3 23/26, RAN4 24/26, RAN5 20/26 (RAN5, scoped to conformance testing, omits Agreement / WorkingAssumption /

Summary / SessionNotes / Chart / Figure). These absences are expected and not counted as failures: they indicate that a non-RAN1 WG does not need to populate the full RAN1 ontology to benefit from reuse.

6.5 E5 — Reusability evidence via public artifacts

To illustrate what the ontology is for, we list a curated set of competency questions that SPECTRA supports against an instantiated knowledge graph; each question is paired with the schema fragment it exercises and the role it plays in the standardization workflow.

- **Multi-hop traceability.** "Which TDoc motivated the current text of TS 38.214 §5.1.3?" — starts at the Section, traverses CRs that modify it (reverse traversal over `modifiesSection`), reaches the Resolutions linked through the indirect join of Figure 1, and finally retrieves the TDocs cited by those Resolutions (reverse traversal over `references`).
- **Cross-WG impact.** "Which RAN2 contributions were triggered by a RAN1 Liaison Statement in meeting M?" — uses `LS.originatedFrom` / `sentTo` between `WorkingGroup` nodes.
- **Release-scoped & lifecycle queries.** "CRs against Release 18 by Company X" combines `CR.targetRelease`, `CR.modifies`, `Tdoc.submittedBy`; "Working Assumptions promoted to Agreements" follows `WorkingAssumption.promotedTo`.

Concretely, the multi-hop traceability query reduces to a Cypher pattern that joins `Section` $\xleftarrow{\texttt{MODIFIES_SECTION}}$ `CR`, the indirect `CRPack/Meeting` link to `Resolution`, and `Resolution` $\xrightarrow{\texttt{REFERENCES}}$ `Tdoc`; the full executable form is shipped in the release as `queries/cypher/`. The release ships executable Cypher translations of 14 representative CQs together with 5 SPARQL example queries, plus an end-to-end synthetic example (`examples/end_to_end/`) where a complete TDoc → Resolution → CR → Section → Spec scenario is bundled with both Cypher and SPARQL queries and the expected output, so that reusers can verify their toolchain against SPECTRA before instantiating it on their own data. The full 137-CQ validation suite is retained as internal validation material (Section 5).

Cross-WG queries on deployed KGs. The same schema across the five RAN per-WG KGs supports queries that cross WG boundaries; counts (Cypher and results in `validation/cross_wg_use_evidence.json`) are measured on the internal KGs. The RAN1 KG records 3,644 LSes to RAN2 (largest recipient; RAN4 1,457, RAN3 380); the RAN2 KG returns 1,450 RAN1-originated LSes (peak: RAN2 #118-e). TS 38.300 is jointly modified by 2,039 RAN2 CRs and 270 RAN3 CRs — all surfaced via SPECTRA's `modifies` without per-WG customisation. SPECTRA also underwrites a standards-engineering knowledge infrastructure at Samsung System LSI.

7 Availability, Sustainability, and FAIRness

We follow a *copyright-aware FAIR release pattern*: schema, validation evidence, and reusable query examples are released openly, while the instantiated KG and parsing pipeline remain internal due to 3GPP TDoc redistribution restrictions. A `validation/` directory of JSON evidence (structural metrics, OOPS!, cross-WG schema diff, cross-WG use-evidence) plus `validation_manifest.md` maps every quantitative claim in this paper to its supporting file, so reviewers can inspect each cited number directly.

Publicly released (CC-BY 4.0).

- **SPECTRA ontology** (OWL 2 Turtle); the ontology declares `https://w3id.org/spectra` as its IRI, to be registered with the W3C Permanent Identifier service before the camera-ready version. **PyLODE HTML documentation** and a **SHACL shapes file** for core cardinality/range constraints.
- **README** (scope, version, license) and machine-readable metadata (`CITATION.cff`, `codemeta.json`); a schema diagram.
- **Representative competency questions** (curated subset of the 137-CQ suite) with 14 **executable Cypher translations** and 5 **SPARQL examples**, plus a **synthetic instantiation snippet**.
- **Validation evidence** (`validation/`): JSON files for structural metrics, OOPS! pitfall scan, cross-WG schema diff, and cross-WG use-evidence counts, each linked to a paper claim by `validation_manifest.md`.

Not publicly released.

- The complete 137-CQ validation suite and the full set of executable Cypher queries used for cumulative regression testing (retained as internal validation material; representative subsets are released as noted above).
- The instantiated 3GPP RAN knowledge graphs (per-WG instantiations at realistic scale; not redistributable because the underlying TDocs are subject to 3GPP copyright; the original TDocs remain publicly accessible via the 3GPP portal at `https://www.3gpp.org`).
- The operational parsing and knowledge-graph population pipeline (internal project code; not released).

FAIR alignment. **F:** a persistent Zenodo DOI will be minted from the tagged GitHub release upon camera-ready publication. **A:** the OWL file and documentation are downloadable without authentication. **I:** OWL 2 Turtle, loadable by standard tooling (Protégé, Jena, RDFLib); reuses DC and FOAF. **R:** CC-BY 4.0, versioned, with README, representative queries, and the validation manifest above.

Sustainability. SPECTRA serves as the semantic layer of a standards-engineering knowledge infrastructure at Samsung System LSI; incremental updates per 3GPP meeting cycle are produced internally, the public release is versioned accordingly, and changes are validated against the cumulative CQ protocol before release.

8 Conclusion

We presented **SPECTRA** as a reusable OWL ontology resource for 3GPP RAN standardization (CC-BY 4.0, Zenodo DOI upon publication), grounded in 137 CQs validated by a five-phase cumulative protocol; the released schema is shown to be cross-WG stable across RAN1–RAN5 with only one additional OP and two genuine DPs needed for RAN2–RAN5. The instantiated KGs and parsing pipeline are retained as internal validation evidence (3GPP copyright, Section 7).

Limitations and future work. Scope is the RAN TSG; extending to TSG SA/CT would add classes and relationships. The released queries are a representative subset of the full 137-CQ executable suite, retained internally with the per-WG KGs and parsing pipeline (3GPP copyright); no public SPARQL endpoint is provided. Adoption evidence is currently confined to the authors' organisation, and CQ correctness was verified by the eliciting engineers themselves. We plan to extend to TSG SA/CT, investigate LLM-assisted CQ generation, examine other standards bodies (IETF, IEEE), and engage the Semantic Web community for independent reuse evidence.

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Resource Availability Statement: The SPECTRA ontology, SHACL shapes, PyLODE HTML documentation, anonymized 137-CQ index, 14 Cypher and 5 SPARQL representative queries, end-to-end synthetic example, and `validation/` directory (with `validation_manifest.md` mapping every quantitative claim to its supporting JSON) are publicly released under CC-BY 4.0 at `https://github.com/spectra-ontology/spec-trace`; a Zenodo DOI minted from the tagged release will replace this URL upon camera-ready publication. The full 137-CQ executable suite, per-WG KGs, and parsing pipeline are retained internally because the underlying 3GPP TDocs are subject to copyright; the TDocs remain publicly accessible via `https://www.3gpp.org`.

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